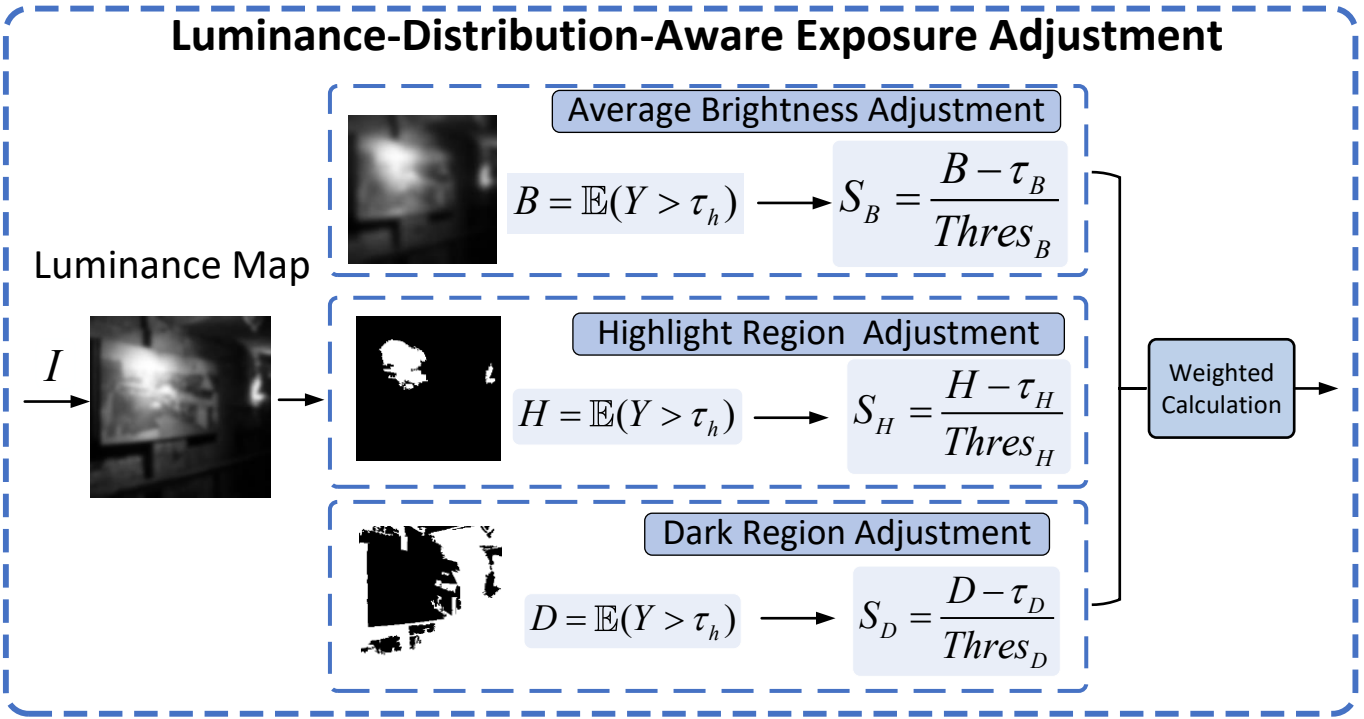
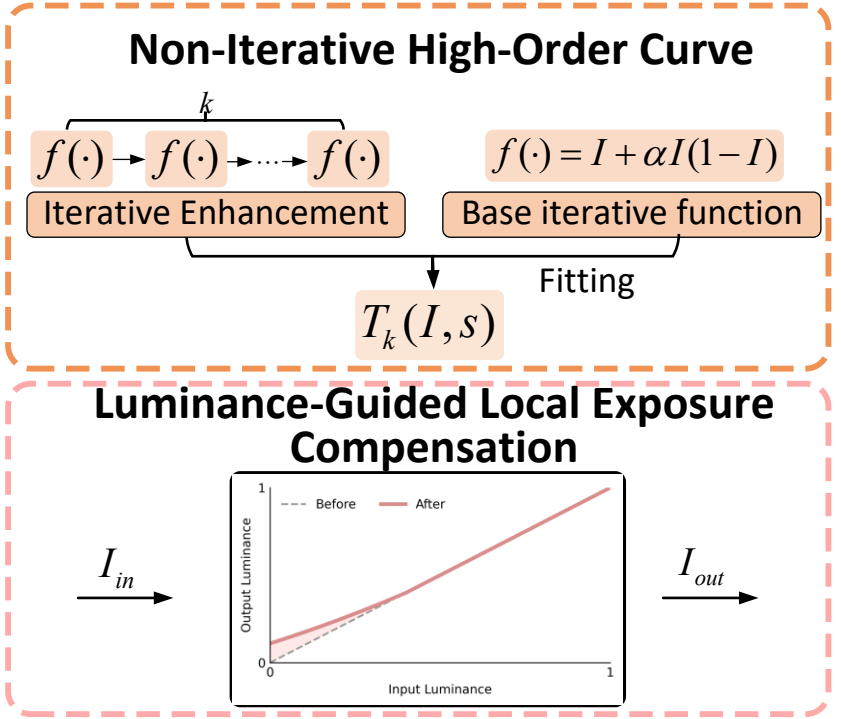

Input
Ultra-Lightweight Exposure Enhancement Network
Conv 3X3 3→1
Tanh
Luminance-Distribution-Aware Exposure Adjustment
Non-Iterative High-Order Curve
Luminance-Guided Local Exposure Compensation
Ouput


Luminance-Distribution-Aware Exposure Adjustment
Luminance Map
I
Average Brightness Adjustment
$B = \mathbb{E}(Y > \tau_h)$
$S_B = \frac{B - \tau_B}{Thres_B}$
Highlight Region Adjustment
$H = \mathbb{E}(Y > \tau_h)$
$S_H = \frac{H - \tau_H}{Thres_H}$
Dark Region Adjustment
$D = \mathbb{E}(Y > \tau_h)$
$S_D = \frac{D - \tau_D}{Thres_D}$
Weighted Calculation


Non-Iterative High-Order Curve
$f(\cdot) \rightarrow f(\cdot) \rightarrow \cdots \rightarrow f(\cdot)$
k
Iterative Enhancement
$f(\cdot) = I + \alpha I(1 - I)$
Base iterative function
Fitting
$T_k(I, s)$
Luminance-Guided Local Exposure Compensation
I_{in}
Before
After
Output Luminance
Input Luminance
I_{out}